PW 3

1 Project Creation & Setup

Just like all **PW** just create an empty views activity project

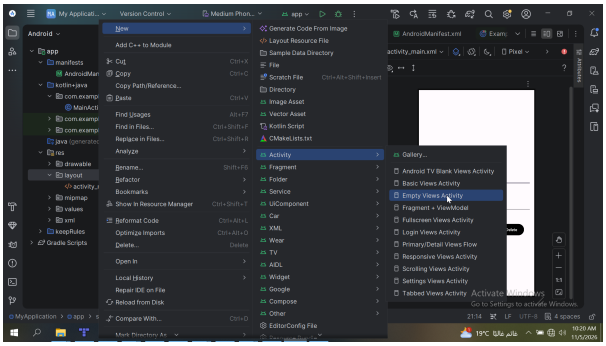
2 Application

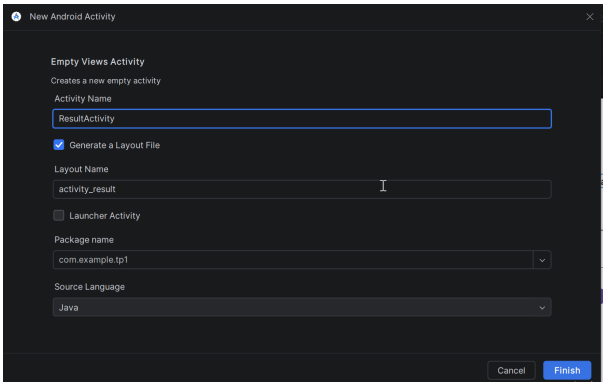
In this **PW** we will create an application with 2 activities :

- **main :** takes height_{cm} and width_{kg} to compute **BMI** $= \dfrac{\text{width}_{kg}}{\text{height}^2_{m}}$ sends it to next activity
- **result :** shows the **BMI** along with the interpretation

2.1 Create Result activity

Left click on main menu : `new > activity > empty views activity` then input name of the activity





2.2 Layout

We're gonna start with the main activity layout then the result activity layout :

2.2.1 Main Activity Layout

Like before we will use **LinearLayout** set it to **vertical** direction :

- **TextView App Title** : My Health Coach
- **TextView Weight** : Weight(Kg)
- **EditText Weight** : Numerical field to input weight in `kg`
- **TextView Height** : Height(cm)
- **EditText Height** : Numerical field to input height in `cm`
- **LinearCompat Button** : We will use another **LinearLayoutCompat** this time set to **horizontal** direction to make buttons right after the other :
  - **Compute Button** : Compute **BMI** and moves to next activity
  - **Delete Button** : Empty all fields

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<!-- we set the orientation to vertical, set the context to MainActivity and layout to LinearLayout
that takes entire screen (math_parent) and alignment to center (gravity) -->
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
        xmlns:app="http://schemas.android.com/apk/res-auto"
        xmlns:tools="http://schemas.android.com/tools"
        android:id="@+id/main"
        android:layout_width="match_parent"
        android:layout_height="match_parent"
        android:gravity="center"
        android:orientation="vertical"
        tools:context=".MainActivity">

<!-- "My Health Coach" bold label with 25sp text size that takes the minimum space (wrap_content)
and has margin bottom of 40dp -->
    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="My Health Coach"
        android:textSize="25sp"
        android:textStyle="bold"
        android:layout_marginBottom="40dp"
        />

<!-- "Weight (Kg)" bold label with 18sp text size
that takes the minimum height (wrap_content)
but takes the entire screen width(match_parent) to be aligned to the left
and has margin bottom/left of 5dp -->
    <TextView
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Weight (Kg)"
        android:textSize="18sp"
        android:textStyle="bold"
        android:layout_marginBottom="5dp"
        android:layout_marginLeft="5dp"
        />

<!-- textfield that only takes decimal number with an id "weightField"
takes the entire screen width(match_parent) but minimum height (wrap_content)
margin left right of 12dp (horizontal)
margin bottom of 20dp
placeholder (hint) = "Ex: 75" -->
    <EditText
        android:id="@+id/weightField"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_marginHorizontal="12dp"
        android:hint="Ex: 75"
        android:inputType="numberDecimal"
        android:layout_marginBottom="20dp"
        />

<!-- "Height (Cm)" bold label with 18sp text size
that takes the minimum height (wrap_content)
but takes the entire screen width(match_parent) to be aligned to the left
and has margin bottom/left of 5dp -->
    <TextView
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Height (Cm)"
        android:textSize="18sp"
        android:textStyle="bold"
        android:layout_marginBottom="5dp"
        android:layout_marginLeft="5dp"
        />
```

activity_main.xml

```
<!-- textfield that only takes decimal number with an id "heightField"
takes the entire screen width(match_parent) but minimum height (wrap_content)
margin left right of 12dp (horizontal)
margin bottom of 20dp
placeholder (hint) = "Ex: 180"-->
    <EditText
        android:id="@+id/heightField"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_marginHorizontal="12dp"
        android:hint="Ex: 180"
        android:inputType="numberDecimal"
        android:layout_marginBottom="20dp"
        />

<!-- we set the orientation to horizontal, layout to LinearLayout
takes the entire screen width(match_parent) but minimum height (wrap_content)
and alignment to center (gravity) -->
    <LinearLayout
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:gravity="center"
            android:layout_marginVertical="20dp"
            android:orientation="horizontal">

<!-- "Compute "button with an id "computeButton"
has width of 250dp but minimum height (wrap_content)
margin right of 10dp -->
        <Button
            android:id="@+id/computeButton"
            android:layout_width="250dp"
            android:layout_height="wrap_content"
            android:text="Compute"
            android:layout_marginRight="10dp"
            />


<!-- "Delete" black button with an id "delButton"
takes the minimum size (wrap_content)
when clicked it will call the method "delete" in the context "MainActivity" -->
        <Button
            android:id="@+id/delButton"
            android:backgroundTint="@color/black"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="Delete"
            android:onClick="delete"/>
    </LinearLayout>


</LinearLayout>
```

2.2.2 Result Activity Layout

We will use **LinearLayout** set it to **vertical** direction :

- **TextView App Title** : My Health Coach
- **TextView BMI** : label that holds computed **BMI**
- **TextView Interpretation** : Interpretation label
- **Back Button** : Moves back to main activity

activity_result.xml

```
<?xml version="1.0" encoding="utf-8"?>
<!-- we set the orientation to vertical, set the context to ResultActivity and layout to LinearLayout
that takes entire screen (math_parent) and alignment to center (gravity) -->
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
        xmlns:app="http://schemas.android.com/apk/res-auto"
        xmlns:tools="http://schemas.android.com/tools"
        android:id="@+id/main"
        android:layout_width="match_parent"
        android:layout_height="match_parent"
        android:gravity="center"
        android:orientation="vertical"
        tools:context=".ResultActivity">

<!-- "My Health Coach" bold label with 25sp text size that takes the minimum space (wrap_content)
and has margin bottom of 40dp -->
    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="My Health Coach"
        android:textSize="25sp"
        android:textStyle="bold"
        android:layout_marginBottom="40dp"/>

<!-- "BMI : " bold label with 25sp text size that takes the minimum space (wrap_content)
with an id of "bmiText"
and has margin bottom of 40dp -->
    <TextView
        android:id="@+id/bmiText"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="BMI : "
        android:textSize="25sp"
        android:textStyle="bold"
        android:layout_marginBottom="40dp" />

<!-- "Interpretation : " bold label with 25sp text size that takes the minimum space (wrap_content)
with an id of "interpretationText"
and has margin bottom of 40dp -->
    <TextView
        android:id="@+id/interpretationText"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Interpretation : "
        android:textSize="25sp"
        android:textStyle="bold"
        android:layout_marginBottom="40dp"/>

<!-- "Back" button with an id "backButton"
takes the entire screen width(match_parent) but minimum height (wrap_content)
margin left right of 40dp (horizontal)
when clicked it will call the method "back" in the context "ResultActivity"
-->
    <Button
        android:id="@+id/backButton"
        android:layout_width="match_parent"
        android:layout_marginHorizontal="40dp"
        android:layout_height="wrap_content"
        android:text="Back"
        android:onClick="back"/>

</LinearLayout>
```

2.3 Context

2.3.1 Main Activity

We need to declare the following attributes and bind them to the xml in `onCreate` :

- **weightField**
- **heightField**
- **computeButton**

and we need to implement the following methods :

- **compute Method**
- **delete Method**

Note that since we didn't define compute in `onClick` we will need to use a listener

MainActivity.java

```java
package com.example.tp3;

import static android.content.ContentValues.TAG;
import android.annotation.SuppressLint;
import android.content.Intent;
import android.os.Bundle;
import android.util.Log;
import android.view.View;
import android.widget.Button;
import android.widget.EditText;
import android.widget.TextView;
import android.widget.Toast;
import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;

public class MainActivity extends AppCompatActivity {

    // declaration of fields and compute button
    private EditText weightField;
    private EditText heightField;
    private Button computeButton;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);

        Log.d(TAG,"onCreate()");
        EdgeToEdge.enable(this);
        setContentView(R.layout.activity_main);

        // binding with xml layout
        weightField = findViewById(R.id.weightField);
        heightField = findViewById(R.id.heightField);
        computeButton = findViewById(R.id.computeButton);


        ViewCompat.setOnApplyWindowInsetsListener(findViewById(R.id.main), (v, insets) -> {
            Insets systemBars = insets.getInsets(WindowInsetsCompat.Type.systemBars());
            v.setPadding(systemBars.left, systemBars.top, systemBars.right, systemBars.bottom);
            return insets;
        });

        // using listener onClick for computeButton
        computeButton.setOnClickListener(this::compute);
    }

    // empties weightField and heightField
    public void delete(View view) {
        weightField.setText("");
        heightField.setText("");
    }
```

ActivityMain.java

```
    public void compute(View view) {
        // if one field is empty create a toast
        String weightValue = weightField.getText().toString().trim();
        String heightValue = heightField.getText().toString().trim();
        if (weightValue.isEmpty() || heightValue.isEmpty()) {
            Toast.makeText(this, "please fill all fields", Toast.LENGTH_LONG).show();
            return;
        }

        // else parse to double and compute BMI
        double weight = Double.parseDouble(weightValue);
        double height = Double.parseDouble(heightValue)/100.;
        height =  Math.pow(height,2);
        double BMI = weight / height;

        // create intent to go to ResultActivity and put the BMI in "EXTRA_BMI"
        Intent path = new Intent(this,ResultActivity.class);
        path.putExtra("EXTRA_BMI",BMI);
        startActivity(path);
    }
}
```

2.3.2 Result Activity

We need to declare the following attributes and bind them to the xml in `onCreate` :

- **bmiText**
- **interpretationText**

and we need to implement the following methods :

- **back Method**
- **onStart Method**

In `onStart` we will fetch the extras.

ResultActivity.java

```java
package com.example.tp3;

import android.content.Intent;
import android.os.Bundle;
import android.view.View;
import android.widget.EditText;
import android.widget.TextView;
import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;

public class ResultActivity extends AppCompatActivity {

    // declaration of the labels
    private TextView bmiText;
    private TextView interpretationText;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        EdgeToEdge.enable(this);
        setContentView(R.layout.activity_result);

        // binding with xml layout
        bmiText = findViewById(R.id.bmiText);
        interpretationText = findViewById(R.id.interpretationText);

        ViewCompat.setOnApplyWindowInsetsListener(findViewById(R.id.main), (v, insets) -> {
            Insets systemBars = insets.getInsets(WindowInsetsCompat.Type.systemBars());
            v.setPadding(systemBars.left, systemBars.top, systemBars.right, systemBars.bottom);
            return insets;
        });

    }

    // fetch the EXTRA_BMI (default value 0), and format it to 2 numbers after float point
    // and set BMI text
    @Override
    protected void onStart() {
        super.onStart();
        double BMI = getIntent().getDoubleExtra("EXTRA_BMI",0.0);
        String BMIFormated = String.format("%.2f",BMI);
        bmiText.setText("BMI : "+BMIFormated);
    }

    // create intent to go back to Main activity
    public void back(View view) {
        Intent path = new Intent(this,MainActivity.class);
        startActivity(path);
    }

}
```

3 Execution

Im still executing the application in my physical android phone below are the screenshot of the execution :

